Natural logarithmic functions

1

a $\frac{dy}{dx} = \frac{1}{x}$

b $\frac{dy}{dx} = \frac{7}{x}$

c $\frac{dy}{dx} = \frac{1}{x}$

d $\frac{dy}{dx} = \frac{2}{x}$

e $\frac{dy}{dx} = \frac{1}{x}$

f $\frac{dy}{dx} = \frac{4}{x} - 1$

g $\frac{dy}{dx} = 4x^3 + \frac{6}{x}$

h $\frac{dy}{dx} = \frac{4}{x} + \frac{1}{x^2}$

i $\frac{dy}{dx} = \frac{8}{x}$

j $\frac{dy}{dx} = \frac{7}{7x+3}$

k $\frac{dy}{dx} = \frac{4x^3}{x^4+2}$

l $\frac{dy}{dx} = \frac{2x+7}{x^2+7x+5}$

m $\frac{dy}{dx} = \frac{2}{x-3}$

n $\frac{dy}{dx} = \frac{4+3x^2}{2+4x+x^3}$

o $\frac{dy}{dx} = \frac{36x^2+48x}{4x^3+8x^2-9}$

p $\frac{dy}{dx} = \frac{1}{x}$

q $\frac{dy}{dx} = \frac{4}{x}$

r $\frac{dy}{dx} = -\frac{1}{x}$

s $\frac{dy}{dx} = \frac{-1}{8-x}$

t $\frac{dy}{dt} = \frac{2t-5}{t^2-5t}$

u $\frac{dy}{dx} = -\frac{1}{x+4}$

v $\frac{dy}{dx} = -\frac{3}{3x+4}$

2

a $y = \ln(u)$

b $\frac{du}{dx} = \frac{6}{(x+3)^2}$

c $\frac{dy}{du} = \frac{x+3}{x-3}$

d $\frac{dy}{dx} = \frac{6}{(x-3)(x+3)}$

3

a $y = 2\ln x$

b $\frac{dy}{dx} = \frac{2}{x}$

c $x = 8$

4

a $y = 4\ln(5x-2)$

b $y' = \frac{20}{5x-2}$

c $x = \frac{62}{5}$

5

a $\frac{dy}{dx} = \frac{1}{2x}$

b $\frac{dy}{dx} = \frac{1}{2x}$

c $\frac{dy}{dx} = -\frac{5}{4-10x}$

6

a $f'(x) = \frac{x}{x^2+1}$

b $f'(2) = \frac{2}{5}$

7

a $f'(x) = \dfrac{32x}{4x^2 + 3}$

b $x = \dfrac{1}{2}$, $x = \dfrac{3}{2}$

Multiple differentiation rules

8

a $\dfrac{dy}{dx} = 1 + \ln x$

b $\dfrac{dy}{dx} = \dfrac{6(\ln x)^5}{x}$

c $\dfrac{dy}{dx} = \dfrac{1 - \ln x}{x^2}$

d $\dfrac{dy}{dx} = \dfrac{\ln x - 1}{(\ln x)^2}$

e $\dfrac{dy}{dx} = 9x^2 \ln x + 3x^2$

f $\dfrac{dy}{dx} = 6x^3 + 24x^3 \ln x$

g $\dfrac{dy}{dx} = 5x^4 \ln(x - 5) + \dfrac{x^5}{x - 5}$

h $\dfrac{dy}{dx} = \ln(x + 6) + 1$

i $\dfrac{dy}{dx} = \dfrac{x^2 + 6}{x} + 2x \ln 2x$

j $\dfrac{dy}{dx} = e^x \ln x + \dfrac{1}{x} e^x$

k $\dfrac{dy}{dx} = 8$

l $\dfrac{dy}{dx} = 1$

m $\dfrac{dy}{dx} = 6e^{6x} \ln 4x + \dfrac{e^{6x}}{x}$

n $\dfrac{dy}{dx} = \dfrac{1}{x \log_e 4x}$

o $\dfrac{dy}{dx} = \dfrac{1 - x \ln x}{xe^x}$

p $\dfrac{dy}{dx} = \dfrac{5x^4 + 2 - 20x^4 \ln 3x}{x(5x^4 + 2)^2}$

9

a $\dfrac{dy}{dx} = 3x^2 \ln x + x^2$

b $13e^8$

10 $\dfrac{d}{dx}(\ln(f(x))) = 4$

11 $f'(2) = \dfrac{9}{4}$

12

a No solution provided.

b $f'(x) = 3xe^{3x} + e^{3x}$

13

a $\dfrac{\frac{1}{x} \times f(x) - f'(x) \times \ln x}{(f(x))^2}$

b $g'(e^3) = -\dfrac{1}{8e^6}$

14

a $f'(x)$ … k

b $g'(x)$ … 1

c k times faster

15 No solution provided.

Bases other than e

16

a $x = 3^y$

b $x = e^{y \ln 3}$

c $\dfrac{dx}{dy} = e^{y \ln 3} \ln 3$

d $\dfrac{dx}{dy} = x \ln 3$

e $\dfrac{dy}{dx} = \dfrac{1}{x \ln 3}$

17

a $f'(x) = \dfrac{1}{x \ln 2}$

b $f'(x) = \dfrac{1}{x \ln 2}$

c $f'(x) = \dfrac{1}{(x-2) \ln 3}$

d $f'(x) = -\dfrac{1}{(1-x) \ln 10}$

e $f'(x) = \dfrac{2}{(2x+1) \ln 5} - 6$

f $f'(x) = -\dfrac{7}{(2-7x) \ln 6}$